

Population-based screening for cardiovascular diseases and type 2 diabetes risk among migrant origin and ethnic minority groups in Europe: a scoping review

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Population-based screening for cardiovascular diseases and type 2 diabetes risk among migrant origin and ethnic minority groups in Europe: a scoping review

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Abstract

Introduction Although some migrant origin groups in Europe show elevated risk for cardiovascular diseases (CVDs) and type 2 diabetes (T2D), information is sparse and fragmented. This study examines availability and possibilities for harmonization of health examination survey (HES) data on major determinants of CVD and T2D among migrant origin and ethnic minority groups in EU Member states and EU4Health associated countries (Norway, Iceland, Ukraine, Moldova, Montenegro).

Methods We conducted a scoping review, following PRISMA-ScR in PubMed and Web of Science, supplemented by targeted grey literature searches of national health institutes, to identify HESs covering core risk factors for CVD and T2D among migrant origin and ethnic minority groups, older than 18 years and living in EU Member states and selected associated countries. Studies were published between 2014 and 2026. Altogether 2537 peer-reviewed records and 348 grey literature reports were screened. Following full-text screening, 57 peer-reviewed papers were included.

Results In total, 24 individual HESs in 10 countries were identified. The majority (n=38, 67%) of the surveys were conducted in the Netherlands or Sweden. None of the HESs used nationally representative samples, having been conducted regionally. There was notable heterogeneity in how persons with a migration background and ethnic minorities were defined and grouped. Reported risk factors included obesity, hypertension, hypercholesterolemia, and hyperglycemia. Measurement methods of these risk factors varied by the migrant origin group and country where the study was conducted.

Conclusion: We did not identify any eligible publications on CVD or T2D risk factors in migrant origin or ethnic minority groups in 69% of EU Member States/selected associated countries, indicating substantial knowledge gaps among these population groups in Europe. The heterogeneity in measurement methods makes cross-country

comparison and data harmonization challenging. Data availability and cross-country comparability should be improved to support effective evidence-based health promotion and prevention measures.

Key words: ethnicity, cardiovascular diseases, type 2 diabetes, risk factors, health examination survey

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Key Messages

What is already known on this topic

- National health examination surveys (HESs) are important tools for monitoring cardiovascular diseases (CVDs) and type 2 diabetes (T2D).

What this study adds

- This study identified major gaps in the availability of HES data on measured risk factors for CVD and T2D among migrant origin and ethnic minority groups and indigenous people across EU Member states and selected associated countries.
- Major methodological heterogeneity was observed in how migrant origin and ethnic minority groups were defined, as well as in how key CVD and T2D risk factors (obesity, hypertension, hypercholesterolemia, and hyperglycemia) were measured.

How this study might affect research, practice or policy

- This study underlines the need for nationally representative data, standardized definitions, and uniform methods to support evidence-based policy and practice and enable cross-country comparisons.
- Based on our findings, we recommend coordinated guidance at the European level on improving data availability and inclusion of migrant origin and ethnic minority groups and indigenous people in population-based screening initiatives.

Introduction

Cardiovascular diseases (CVDs) and type 2 diabetes (T2D) are among the leading causes of morbidity and mortality globally [1]. Common modifiable behavioral risk factors include smoking, excess alcohol use, unhealthy diet, and low physical activity [2,3]. Moreover, physiological risk factors encompass high systolic blood pressure (SBP), high body-mass index (BMI), and high low-density lipoprotein (LDL) cholesterol levels [3]. Diabetes is also a common comorbidity of CVDs, as well as a major risk factor for CVDs, resulting in higher CVD morbidity and mortality in people with T2D [4]. Early identification of high-risk individuals for preventive care is essential [5].

In recent decades, high-income countries (HICs) in Europe have become increasingly culturally and ethnically diverse. As a result, societies now include longstanding ethnic minorities, indigenous people, and a growing population of migrant origin, reshaping demographics, and social structures [6]. In this review, we define ethnic minorities as individuals of migrant origin as well as other cultural and linguistic minority groups, such as the Roma population. Indigenous people refer to populations that have inhabited the given country or region at the time of conquest, colonization, or establishment of current state borders. In the European Union (EU), the Sámi people are the only indigenous group [7]. Migrant-origin groups include individuals who have migrated across national borders, as well as their descendants. Although there has been significant progress in CVD and T2D risk management, diagnostics and treatment across Europe over recent decades, the benefits have not been distributed equally across different population groups [3]. Studies have shown significant variations in CVD and T2D incidence, prevalence, and mortality between persons of migrant origin and ethnic minority groups, and indigenous people as compared to majority populations. For example, stroke is more prevalent among people of sub-Saharan African origin than among majority populations [8], and T2D occurs 10–20 years earlier in many migrant-origin groups [9]. Migrant origin groups also face a higher risk of delayed diagnosis,

contributing to more severe disease progression [10]. To address these inequities, more detailed information on the distribution of risk factors within and between ethnic minority groups are needed. Such data support better-targeted prevention strategies, resource allocation, and equitable healthcare provision [11]. Robust and inclusive screening initiatives are also essential for early detection and improved prevention [12].

National health examination surveys (HESs) provide key information on the prevalence of risk factors for CVD and T2D, essential for identifying high-risk groups and planning health promotion measures both at risk group and population level [13]. Over the past decade, 12 EU Member States, including the UK, have conducted national HESs to collect essential objective data, such as blood pressure, anthropometric measurements and biological samples, complementing self-reported information from health interviews and registers [13–15]. These surveys are considered highly externally valid due to their randomized, representative sampling methods [16]. Compared to self-reported data, HESs substantially reduce reporting bias and measurement error by applying standardized protocols and calibrated instruments, thereby improving reliability and comparability across populations and over time [15,16]. The inclusion of biological samples further enhances the capacity to identify biomarkers and undiagnosed conditions, positioning HESs as a methodological gold standard for population health monitoring [17].

Despite their strengths, existing HESs vary in terms of target populations and types of measurements [18], and most do not systematically include migrant origin and ethnic minority groups, who are often at higher risk for CVD and T2D. This exclusion limits the ability to monitor health disparities and develop equitable prevention strategies. Moreover, there is a lack of research critically examining the methodological challenges of using HES data in this context, particularly regarding data availability, representativeness, and cross-country comparability. These gaps hinder a nuanced

understanding of risk factor distribution and challenge efforts to formulate cross-nationally consistent public health responses.

Recent reviews have examined cardiometabolic health among migrant and ethnic minority populations in Europe; however, they have not evaluated the availability, quality, or comparability of data needed to monitor these disparities. Earlier work pointed to major gaps and inconsistencies in ethnically relevant CVD and diabetes data, including variation in how ethnicity and health outcomes were defined and recorded [19]. More recent reviews have summarized cardiometabolic patterns among migrant populations, yet they do not assess the underlying data infrastructure, nor whether existing data sources can support cross-country comparisons [11,20–22]. Therefore, the aim of this scoping review is to identify and describe studies that have explored the use of HES data on key risk factors for CVD and T2D among migrant-origin, ethnic minority, and indigenous people in Europe.

Methods

Design

We carried out a scoping review following Levac and colleagues' update of Arksey & O'Malley's methodological framework [23]. This framework includes five main stages: (1) identifying the research questions; (2) identifying relevant studies; (3) study selection; (4) charting the data; and (5) collecting, summarizing, and reporting results. We followed the PRISMA-ScR guidelines [24].

Identifying the Research Questions

The following research questions were identified: (1) Which countries in Europe have conducted a HES covering key risk factors for CVD and T2D among migrant origin and ethnic minority groups, and indigenous people? In this review, we defined Europe as the EU Member States and the EU4Health-associated countries (Norway, Iceland, Ukraine,

Moldova, Montenegro), as HESs are predominantly implemented and coordinated within the EU and EU4Health framework; (2) How are these HESs designed?; (3) How are migrant origin and ethnic minority groups, and indigenous people identified within these HESs; and (4) How is data collected within these HESs?.

Search strategy and Eligibility Criteria: Identifying Relevant Studies

Literature searches were conducted in PubMed and Web of Science on September 2, 2024, and updated on February 12, 2026, according to a predetermined search strategy (see Additional file 1). We used Boolean operators and modified our search strategy according to the database. We used a combination of keywords for 'migrant', 'immigrant' and 'ethnic minority', as well as search terms for selected risk factors for CVD and T2D (i.e. obesity, hypertension, hypercholesterolemia, and hyperglycemia). Additional file 2 provides an overview of all keywords and search terms used in this scoping review. We anticipated that the term 'health examination survey' may only be discussed in the full-text of articles, and that other terms may have been used to describe it. As such, this keyword was excluded from our search strategy to allow for a more thorough search of all the studies published in the field. Consequently, this information was assessed during the title, abstract, and full-text screening.

To reduce publication bias and identify relevant national health reports not indexed in bibliographic databases, we extended our search to grey literature on February 12, 2026. We conducted targeted website searches of National Public Health Institutes and Ministries of Health across the included European countries (e.g. Robert Koch Institute in Germany, Sciensano in Belgium, Finnish Institute for Health and Welfare (THL) in Finland). These searches were performed using Google Advanced Search operators (e.g. *site:gov.xx* and *filetype: pdf*) combined with native-language keywords for migrants and selected risk factors (see Additional file 1). To minimize the influence of personalized search algorithms and ensure reproducibility, these searches were performed using a

computer with a clean browser (deleted cookies and history) and without being logged into a Google account. Additionally, to manage potentially overwhelming volume of search engine results, we screened a maximum of 10 pages (approximately 100 hits) for each search combination.

Studies and grey literature files were included if they:

- reported outcomes measured within an HES
- used at least one of the core physical measurements and biological samples of national HESs (i.e. height, weight, blood pressure, and blood samples)
- investigated at least one of the selected risk factors for CVD and T2D, that were obesity, hypertension, hypercholesterolemia, and hyperglycemia
- were based in at least one of the 27 European Member States or the associated non-EU countries of the EU4Health Programme (Norway, Iceland, Ukraine, Moldova, Montenegro)
- were specifically designed to assess risk factors among migrant origin and ethnic minority groups or indigenous people
- included participants older than 18 years old and living in Europe
- were in English or one of the official languages of the countries included
- were published between 2014 and 2026.

Studies and grey literature files were excluded if they:

- involved participants already diagnosed with CVD or T2D
- focused on gestational diabetes
- reported on outcomes measured solely through interviews or register data
- did not have a reference group of majority population in the same region or country of origin

The above exclusion criteria were applied to maintain focus on prevention and risk factors among migrant origin and ethnic minority groups or indigenous people within

comparable contexts. To ensure comparability within a common HES context, we included only studies that reported data for migrant origin, ethnic minority, or Indigenous groups alongside a majority-population reference group from the same region and HES. An internal comparator allowed assessment of whether sampling strategies, measurement procedures, and fieldwork protocols were applied consistently across groups, thereby reducing confounding from cross-survey or cross-regional variation. Without such a reference, observed differences may reflect methodological or contextual differences rather than true group-specific patterns. This criterion therefore ensured that included studies provided a meaningful basis for comparison and allowed the evaluation of whether HESs were designed to capture minority populations relative to the local majority groups. In addition, we concentrated specifically on physiological risk factors for CVD and T2D, as these indicators are measured objectively and consistently within Health Examination Survey (HES) frameworks. HES studies provide standardized biometric data (such as blood pressure, lipid levels, glucose metabolism markers, and anthropometric measurements) which enable reliable cross study comparison and reduce heterogeneity. Focusing on risk factors that HES data can assess robustly also minimized variability arising from self-reported behavioral or psychosocial measures, which are less comparable across countries and surveys.

Study selection

All database search results were imported in Zotero and deduplicated before screening. The remaining references were then uploaded into Rayyan QCRI, a web-tool facilitating collaborations on systematic and scoping reviews [25]. Two reviewers (SMHL and FM) independently screened all titles and abstracts against the inclusion and exclusion criteria. The full texts of the remaining studies were screened based on the eligibility criteria by SMHL and FM. Differences in interpretations were resolved by discussion between the reviewers. Hand searches were conducted after the initial literature search to identify additional relevant studies. This process involved systematically reviewing

the reference lists of all studies that met the inclusion criteria and screening these citations to identify additional eligible studies. Relevant studies identified through this manual process were subsequently included following the same criteria. For grey literature files, we first manually assessed titles, executive summaries, and table of contents to determine if the document potentially constituted a HES relevant to our target group. Documents deemed potentially relevant were then retrieved for full screening. This full-text assessment employed a two-step verification process. Given the complexity and length of national health reports, the initial screening of these documents was facilitated by an AI-powered document analysis tool (NotebookLM, Google). Specific prompts were used to screen the full-texts in accordance with the eligibility criteria. Documents flagged as potentially relevant by the AI tool were then manually verified in full by two reviewers (SMHL and FM) to confirm inclusion. Any uncertainty was resolved through discussion between the reviewers.

Charting the data

The two reviewers (SMHL and FM) independently extracted the data of the included records using a standardized data extraction spreadsheet in Microsoft Excel (see Additional file 3). Both reviewers assessed the complete data extraction spreadsheet, and any disagreement was resolved through discussion between them.

Collating, summarizing, and reporting the results

The synthesis of the charted data was conducted on two levels. Initially, a numerical analysis was performed to assess the scope, characteristics, and distribution of the studies included in the review. This involved generating tables and charts to visualize the distribution of studies by geographic location, study design, migrant origin and ethnic minority groups, indigenous people, and majority populations included in the studies. These insights guided the identification of key research gaps. Subsequently,

the textual data, organized in Excel, were thematically categorized based on risk factors for CVD and T2D.

Results

The database search yielded a total of 2400 publications. After removing duplicates, 2062 titles and abstracts were screened leaving 70 articles for full-text screening. A total of 49 publications that met the eligibility criteria were included. Additionally, hand searches resulted in the further inclusion of six publications. The update of our database search yielded 534 publications. A total of 475 studies were screened by title and abstract, and 11 full-texts were screened. The updated search resulted in the inclusion of two publications. Finally, the extended grey literature search identified 348 records from targeted websites. However, following screening, no grey literature reports were included. The main reasons for exclusion were incorrect outcomes, lack of migrant origin groups/ethnic minorities inclusion, or not meeting the criteria of a HES. Thus, the complete search resulted in a final total of 57 peer-reviewed articles included in the scoping review [26–82]. Figure 1 presents the PRISMA flowchart.

Figure 1. here.

Figure 1: PRISMA flowchart of the search and review process

Overview of health examination surveys in EU Member States and associated countries and study characteristics

In total, data from 24 individual HESs was used. The HESs were spread throughout ten different countries (see Figure 2). Most of the studies were conducted in the Netherlands (n=17), using The Healthy Life in an Urban Setting (HELIUS) study [26–40], the SUNSET study [41], or the Amsterdam Health Monitor study [42]. Twelve studies were conducted in Sweden, using data from the impact of Migration and Ethnicity on Diabetes In Malmö (MEDIM) study [52–59], the 4D Diabetes Project [61,62] and two other cohorts [60].

Other studies were conducted in Norway ($n = 5$), using the SAMINOR 1 and 2 studies [64–67] and the cohort of Norway [68]; Finland ($n = 4$), using data from the Migrant Health and Wellbeing (Maamu) study [69–72]; Italy ($n = 3$), using the Chinese in Prato (CHIP) study [74,75] and another cohort study [73]; and Slovakia ($n = 3$), using the HepaMeta study [76–78]. One cohort study in each following country was conducted: Estonia [79], Portugal [80], Bulgaria [81], and Germany [82]. Moreover, 9 studies used data from the Research on Obesity and Diabetes among African Migrants (RODAM) study which was conducted in the Netherlands, Germany, UK, and Ghana [43–51]. None of the HESs employed nationally representative samples, as all were regionally conducted in recurring locations.

Figure 2. here.

Figure 2: Overview of the EU Member States and associated countries ($n=10$) using data from HES studies ($n=24$)

The participant characteristics varied widely across the included studies, with sample sizes ranging from small groups (e.g., 60 Roma individuals in selected settlements in Bulgaria [81]) to large cohorts e.g., over 3,000 Ghanaian participants across Europe [43,48], or over 4,000 Sámi people in Norway [67]. Gender distribution and mean age were diverse, with male participation being between 30-60%, and average age ranging from 30 to 60 years. An overview of the study and participant characteristics is presented in Table 1, and in more detail in Additional file 4.

Inclusion of people of migrant origin and ethnic minority groups, and indigenous people

The studies were focusing either on persons of migrant origin ($n=49$) [26–63,68–75,79,80,82], ethnic minority groups ($n = 4$) [76–78,81], and indigenous people ($n=4$) [64–67]. Eight studies included migrants from broad regions: the European Union [82], Central-Eastern Europe [60,73], Southern and Western Europe [60], North America [60],

Central and Southern America [73], Africa [61,62], North Africa and Sub-Saharan Africa [73], the Middle-East [60-62], South Asia [68], Asia [61,62,73], or 'other country' [60,79,82]. On the other hand, 41 studies defined more precisely the group to be included: African Surinamese [26-32,35-41], South Asian Surinamese [26-38,41], Somalian [69-72], Ghanaian [26-40,43-51], Cape Verdean [80], Moroccan [26-38,42], Turkish [26-38,42,63], Kurdish [69-72], Iraqi [52-59,63], Chinese [74,75], Finnish [60], Danish [60], or Russian [69-72,79,82]. In one study, participants also had the option to not specify their migration background [82]. The ethnic minority groups that were included across four studies were Roma [76-78,81]. The indigenous Sámi people were also included in four studies [64-67].

Migrant origin and ethnic minority groups, and indigenous people were defined differently across the studies. Eighteen solely included first generation migrants, i.e. being born abroad with at least one parent being born abroad [52-59,68-75,80,82]. Nineteen studies included first- and second-generation migrants (i.e. being born in the country of residence with both parents being born abroad) [26-40,42,61-63]. Additionally, some studies had extra criteria, such as a minimum length of residence in the country of migration to be eligible to participate in the study ($n=5$) [69-72,80], or the requirement of speaking their mother tongue ($n=1$) [82]. Eighteen studies asked participants to report whether they identified as migrants (i.e. self-identification) and used it to define migrant origin groups either as the sole criterion [41] or as a complement to one of the criteria explained above [26-40,42,63]. Finally, two studies did not describe how they defined migrant origin groups [60,79]. For ethnic minority groups, three studies identified the study population through local community workers [76-78]. For indigenous people, four studies used self-reported ethnic background, in combination with the requirement of speaking the language by themselves, or their parent or grandparent [64-67]. Migrant origin or ethnic minority groups, or indigenous

people were compared to either the majority population (n=47) [26-42,52-79,81,82] or the population in the country of origin (n=10) [43-51,80].

Table 1. here.

Measurements of risk factors for CVDs and T2D

In total, 33 studies investigated all risk factors targeted within this review, i.e. obesity, hypertension, hypercholesterolemia, and hyperglycemia [26-28,30,32-35,41-43,45-47,50-54,56-59,63,66,69,70,74-77,79,82]. Additionally, 13 studies examined three out of the four risk factors [29,31,36,37,44,49,55,61,62,67,68,73,81], whereas nine studies examined two risk factors [38-40,48,64,65,70,71,78]. Two studies investigated a single risk factor [60,80]. These risk factors were measured and defined using different methods, explained in more detail below, and in Figure 3 and Table 2. Additional file 5 provides a summary of all risk factors and methods used per study.

Figure 3. here.

Figure 3: Overview of the collected data to measure the selected risk factors

Obesity

Obesity was investigated in 89.5% (n=51) of studies [26-28,30,32-61,63-67,69-77,79,81,82] with different types of anthropometric data, which are shown in Figure 3. The most common data collected were weight [26-28,30,32-38,41-61,63-65,69-77,79] and height [26-28,30,32-38,41-61,63-65,69-77,79,82] in 45 and 46 studies respectively, followed by waist circumference (WC) in 40 studies [28,32,33,36-39,41,43-50,52-61,63-67,70-72,74-77,79,82]. Hip circumference was measured in 20 studies [28,35,37-39,41,44-49,60,70-72,74-77]. A few studies also measured body composition (n=5) [35,38,39,60,82], the sagittal abdominal diameter (n=1) [60] and arm, thigh, and calf circumference (n=1) [35]. Two studies did not describe which anthropometric data was collected [40,81]. Obesity was defined using eight different

indicators, the body mass index (BMI), being the most used. Table 2 provides an overview of the range of cut-off values used to interpret these indicators and assess obesity in the included studies.

Hypertension

Blood pressure was measured in 80.7% (n=46) of the studies [26-37,40-47,50-59,61-63,66-70,73-77,79,80,82]. One study additionally measured resting heart rate [41]. Hypertension was defined at least by one of the following factors (see Figure 3): systolic blood pressure (SBP) (n=29) [26-28,36,37,40-42,44,50,52-59,67-70,73-77,79,82], diastolic blood pressure (DBP) (n=29) [26-28,36,37,40-42,44,50,52-59,67-70,73-77,79,82], being on antihypertensive therapy (n=21) [26-28,36,37,40-42,44,50,58,59,67-70,73-75,79,82], or having a previous diagnosis of hypertension (n=12) [42,52-59,73-75]. However, 17 studies did not describe the criteria used to define hypertension [29-35,43,45-47,51,61-63,66,80]. Table 2 provides the detailed cut-off values mentioned in different studies to assess hypertension.

Hypercholesterolemia

Hypercholesterolemia was measured in 73.7% (n=42) of the studies [26-35,42,43,45-47,49-54,56-59,61-63,66-70,73-79,81,82]. Most of the studies measured cholesterol from fasting blood samples (n=35) [26-35,43,45-47,49-54,56-59,62,69,70,74-79,81,82], while five studies used non-fasting blood samples [42,66-68,73] and two did not specify the type of the blood sample used [61,63]. The length of the fast was reported either lasting overnight (n=11) [30,31,34,50,74-79,81], or between 8 to 12 hours (n=11) [27,28,43,45,49,52-54,56,57,70]. Some studies did not specify the length of the fast (n=13) [26,29,32,33,35,46,47,51,58,59,62,69,82]. Venous blood samples were used in 14 studies [27,43,47,49,51,61,68-70,76-79,81] and capillary blood samples in two studies [74,75]. Blood serum was used for laboratory analysis in six

studies [52,59,62,66–68] and plasma in 13 studies [26,28,35,49,50,53,54,56–58,63,76,77].

Hypercholesterolemia was identified by measuring the concentrations of total cholesterol (TC) [26–32,34,35,42,43,45–47,49–54,56,57,59,61–63,66–70,73–79,81,82], high density lipoprotein (HDL) [27–35,42,45–47,49,50,52–54,56–59,61–63,66–70,76–79,81,82], low density lipoprotein (LDL) cholesterol [27,28,30,35,45,46,49,52–54,56–59,62,63,69,70,73,76–79,81,82], and triglycerides (TG) [28,30–33,35,46,49,53,54,56–58,62,63,66–70,74–79,81,82]. A few studies also measured lipoprotein(a) [79], apolipoprotein A-1 [79], and apolipoprotein B-100 [76,77,79]. Furthermore, 18 studies reported the use of lipid-lowering medication [28–34,42,46,51,53,58,59,68,76–79] and one diagnosis of hypercholesterolemia [79]. Figure 3 provides a visual overview of the data collected. Many studies did not indicate which thresholds were used to define hypercholesterolemia. For those that did report thresholds, Table 2 provides an overview of the wide variety of cut-off values used for defining hypercholesterolemia.

Hyperglycemia

Hyperglycemia was measured in 87.7% (n=50) of studies [26–39,41–59,62–66,68–72,74–79,81,82]. Most studies reported the use of fasting blood samples (n=44) [26–36,38,39,41,43–59,63,69–72,74–79,81,82], whereas five studies used non-fasting blood samples [42,64–66,68]. Moreover, two studies did not report whether the blood samples were drawn from fasting or non-fasting samples [37,62]. The length of the fast was reported either lasting between 8 to 14 hours (n=18) [27,28,36,43–45,48,49,52–58,70–72], or overnight (n=12) [30,31,34,50,63,74–79,81]. The remaining studies did not specify the length of the fast (n=14) [26,29,32,33,35,38,39,41,46,47,51,59,69,82]. Venous blood samples were used in 28 studies [27,38,39,43,44,47–49,51,54–59,63–65,68–72,76–79,81] and capillary blood samples were used in six studies [52,53,62,63,74,75]. Other studies did not specify the origin of the blood sample. Serum

was analyzed in 11 studies [52-55,59,68,70-72,78,81] and plasma in 26 studies [26,28,35,36,39,41,46-51,53-58,63-65,70-72,76,77].

Many different measures and cut-off values were used across studies in defining hyperglycemia (which were used as proxy for T2D) (see Table 2). Moreover, Figure 3 provides a visual overview of the methods used. Hyperglycemia was identified by measuring blood glucose concentration [26-36,38,39,41-59,63-65,68-72,74-79,81,82], performing an oral glucose tolerance test (OGTT) [52-59,62,63,82], or measuring glycated hemoglobin (HbA1c) [28,35,38,39,41,42,46-48,52-54,56,59,62,63,65,66,70-72,82]. Finally, previous diagnosis of T2D and use of medication (including insulin) were also reported in 40 [27,29-31,34,35,37-39,41-48,50,52-59,62-66,68-72,76-79] and 32 studies [27-29,31-39,42-48,50,53,55-59,69,71,72,74,75,82], respectively.

Table 2. Overview of indicators and cut-off values for obesity, hypertension, hypercholesterolemia, and hyperglycemia

Definition method	Cut-off values	N	References
Obesity			
Body Mass Index	25 to ≤ 30 kg/m ² (overweight), ≥ 30 kg/m ² (obesity)	45	[26-28,30,33,34,36-38,40-61,64,65,69-77,79,81,82]
Waist Circumference	≥ 102 cm (male (m)), ≥ 88 cm (female (f))	10	[36,41,43,52,66,67,70-72,82]
	≥ 94 cm (m), ≥ 80 cm (f)	10	[38,47,53-59,79]
Waist-to-hip ratio	≥ 0.9 (m), ≥ 0.85 (f)	19	[28,37-39,44,46-49,60,64,65,70-72,74-77]
Waist-to-height ratio	0.50-0.59 (overweight), ≥ 0.60 (obesity)	8	[33,47,49,70-72,74,75]
Body fat percentage	$\geq 25\%$ (m), $\geq 35\%$ (f)	1	[39]
Relative fat mass	N/A	2	[47,49]
Body shape index	N/A	2	[47,49]
Waist-hip-height ratio	N/A	1	[60]
Hypertension			
Systolic blood pressure	≥ 140 mmHg	29	[26-28,36,37,40-42,44,50,52-59,67-70,73-77,79,82]

Diastolic blood pressure	≥90 mmHg	29	[26-28,36,37,40-42,44,50,52-59,67-70,73-77,79,82]
Hypercholesterolemia			
Total cholesterol (TC)	≥5.0 mmol/L	3	[49,70,79]
	>5.2 mmol/L (≥200 mg/dL)	3	[81]
	≥6.2 mmol/L (≥240 mg/dL)	2	[74,75]
High Density Lipoprotein (HDL)	1.16-1.54 mmol/L (m), 1.29-1.54 mmol/L (f)	1	[62]
	<1.03 mmol/L (m), <1.29 mmol/L (f) (<40 mg/dL (m), <50 mg/dL (f))	8	[76-79]
	<1.0 mmol/L (m), <1.3 mmol/L (f)	1	[53]
	<1.0 mmol/L (m), <1.2 mmol/L (f)	2	[49,70]
Triglyceride (TG)	≥1.7 mmol/L (≥150mg/dL)	13	[32,33,46]
	≥2.26 mmol/L (≥200 mg/dL)	2	[74,75]
Low Density Lipoprotein (LDL)	>2.5 mmol/L	1	[62]
	2.59-4.13 mmol/L (m & f)	1	[62]
	≥3 mmol/L	2	[49,79]
	≥160 mg/dL	1	[46]
TC/HDL-C ratio	>5	3	[42,50,81]
TG/HDL-C ratio	-	2	[49,81]
Apolipoprotein A-1	<1.04 g/L (m), <1.08 g/L (f)	1	[79]
Apolipoprotein B-100	>1.33 g/L (m), >1.17 g/L (f)	1	[79]
Lipoprotein (a)	>0.3 g/L	1	[79]
Hyperglycemia			
Normal Glucose Tolerance	<6.1 mmol/L	2	[54,56]
Fasting plasma glucose (FPG)	≥ 7.0 mmol/L (≥126 mg/dL)	31	[27-29,31,34-36,38,39,41,43-48,50,52-59,69,72,74,75,79,82]
Impaired fasting glucose (IFG)	≥5.6 mmol/L (100 mg/dL)	6	[32,33,53,76-78]
	5.6-6.9 mmol/L	2	[43,46]
	5.7-6.9 mmol/L	1	[41]
	>6.0 mmol/L	2	[70,71]
	>6.0 to <7.0 mmol/L	1	[63]
	≥6.1 mmol/L	2	[81,82]
	≥ 6.1 to < 7.0 mmol/L	7	[52-56,58,59]
Non-fasting blood glucose concentration	>11.0 mmol/L	1	[42]
	≥7.5 mmol/L	2	[64,65]
Non-fasting blood glucose concentration	2-h plasma glucose ≥ 11.1 mmol/L	9	[52-56,58,59,63,82]
Impaired glucose tolerance (IGT)	<i>2-h plasma glucose</i>		
	≥ 7.8 and <11.1 mmol/L	8	[52-56,58,59,63]
	≥8.9mmol/L <12.2mmol/L	1	[63]

	<i>Fasting plasma glucose</i>		
	<7.0 mmol/L	3	[52,58,63]
	<6.1 mmol/L	4	[54-56,59]
Glycated hemoglobin (HbA1c)	≥6.5% (≥48 mmol/mol)	8	[28,39,42,47,65,66,72,82]
	<7 % (53 mmol/mol)	1	[46]
	≥6 % (≥42 mmol/mol)	2	[70,71]

Discussion

In this scoping review, we examined 57 articles and identified 24 HESs covering key risk factors for CVD and T2D among migrant origin and ethnic minority groups in ten European countries (the Netherlands, Sweden, Norway, Finland, Italy, Slovakia, Estonia, Portugal, Bulgaria, and Germany). All of these HES were regional, and no nationally representative HESs were identified. This regional concentration of HESs targeting migrant origin and ethnic minority groups likely reflects their urban clustering, enabling efficient survey logistics but potentially compromising national representativeness.

Design, populations studied, and definitions of migrant origin and ethnic minority groups, and indigenous people varied across the HES. We also found substantial heterogeneity in whether included key risk factors for CVD and T2D (obesity, hypertension, hypercholesterolemia, and hyperglycemia) were assessed and how these were measured and defined, which complicates comparability across studies. Despite the clear public health importance of CVD and T2D, our findings suggest a lack of standardized, inclusive, and coordinated efforts to monitor these conditions among migrant origin and ethnic minority groups, and indigenous people at the national level.

Fragmented HES efforts among migrant origin and ethnic minority groups

Previous research has shown the crucial role of HESs in generating high-quality data that supports early detection, prevention strategies, and informed policymaking at the national level [16,17,83]. However, as many as 67% (n=38) of the included studies were conducted in the Netherlands and Sweden, including the largest scale Dutch studies

HELIUS and RODAM. In contrast, many other European countries had conducted only a single HES involving persons of migrant origin or ethnic minority groups, or none. This uneven distribution highlights significant geographic disparities in the implementation of population-based screening efforts. Notably, countries such as Germany, France, or Spain host some of the largest and most diverse migrant populations in Europe, reflecting broader demographic shifts across the continent [84]. To address this imbalance, future public health research should prioritize expanding HES coverage in underrepresented regions, particularly those with substantial migrant origin and ethnic minority groups. This includes allocating funding, building local research capacity, and fostering cross-country collaboration to ensure that health data collection reflects the diversity of Europe's population.

Inconsistent definitions of migrant origin groups and ethnicity

A key finding of this review is the significant variation in how migrant origin, ethnic minority groups, and indigenous populations are defined and identified across HESs. This reflects both a lack of conceptual and methodological standardization across Europe and differences in the availability of migration and ethnicity related data in population registers commonly used for sampling. Such variation is further shaped by national legal frameworks, which determine what types of data can be collected and accessed for these purposes. Definitions ranged from country of birth and self-reported status to administrative data and community-based identification. Some studies also lacked clarity on how migrant origin groups or ethnicity were defined. This variability reflects broader challenges documented in the literature. Different data sources, such as country of birth, self-identified ethnicity, language, or religious beliefs, are often used as proxies for migrant origin groups, but each comes with important limitations. For instance, country of birth becomes an increasingly crude measure over time, particularly for second- or third-generation migrants, while nationality or citizenship may not reflect cultural identity or health-related behaviors [85–88].

Ethnicity itself implies shared origins, culture, customs, language, or religion [89], yet these dimensions are difficult to capture consistently and meaningfully in HESs. Moreover, the relationship between migrant origin groups and disease risk is complex and shaped by multiple factors across the life course. These include genetic predispositions, early-life exposure to poverty, malnutrition, or infection, cultural practices maintained after migration, and the quality of care and structural conditions encountered in the host countries [6,8]. Second-generation individuals (i.e. those born in the host country to migrant parents) can still face elevated risks for T2D and CVDs [11]. This raises important questions about how long and on what basis people should be classified as belonging to a migrant origin group. Researchers increasingly call for more reflexive, inclusive approaches that avoid rigid stigmatizing categories and instead reflect lived experiences and structural factors [90,91]. Nevertheless, without a shared understanding of what constitutes 'migrant origin' or 'ethnicity', it becomes difficult to interpret differences between population groups.

Heterogeneity in risk factor measurement and diagnostic criteria

There was substantial heterogeneity in the methods used to measure and define the four selected CVD and T2D risk factors (i.e. obesity, hypertension, hypercholesterolemia, and hyperglycemia). Studies varied in indicators, diagnostic thresholds, and sample types, leading to a lack of consistency and transparency. These methodological differences can result in risk misestimation, missed diagnoses, or inappropriate follow-up, potentially leading to unequal access to preventive care, delayed treatment, and inefficient resource allocation. Previous studies have highlighted the need for harmonization of measurement protocols in HESs to enable valid cross-country comparisons and support coordinated public health action at the European level [17,83,92]. Our findings reaffirm this need but also extend it by highlighting a critical gap: existing calls for harmonization rarely address how

standardized methods should be adapted or applied in the context of population diversity.

Implications and future directions

It is crucial that HESs reflect the full diversity of the population, including persons of migrant origin and ethnic minority groups, who are often underrepresented in routine data systems [85]. Given that these populations may comprise a small share of national sampling frames, representativeness requires deliberate design choices, including targeted oversampling or dedicated supplementary sampling. In this regard, migrant origin ethnic minority groups, and indigenous people should be integrated as a supplementary sample in national HESs, rather than in smaller regional HESs specifically focused on these groups. The supplementary sample has to be large enough to conduct meaningful sub-group analyses. Such an approach could increase analytic power and improve the precision of subgroup estimates across Europe. Importantly, it must be kept in mind that one-size-fits-all approach does not equal to successful integration. There should be flexibility acknowledging diverse knowledge systems and cultural norms when designing the questionnaires and health examination protocols. Potential barriers to participation should be considered, and recruitment methods may need to be adapted to support the participation rate. Thus, our findings underscore the urgent need for coordinated guidance at the European level on how to include those people in population-based screening initiatives.

This review shows the inconsistent identification, inclusion of migrant origin or ethnic minority groups, and risk factors assessment across Europe, adding complexity to this issue. While harmonizing definitions and measurement protocols is essential, it is equally important to develop practical guidelines for the inclusion of migration background and ethnicity in HESs. Such guidance should specify which dimensions of diversity need to be examined, provide methodological recommendations on how to

sample for diversity within HESs, and establish consistent definitions for cross-country minorities, such as those of 'migrant origin' or 'ethnic minorities'. Including these dimensions in HESs is not only justified by the known disparities in CVD and T2D risk [6], but also necessary for designing effective, equitable interventions.

Future research should examine how HESs can further serve as a general tool to identify gaps in healthcare, thereby informing the design of targeted approaches to reach high-risk groups. Ultimately, greater consistency in representing migrant origin and ethnic minority groups in health monitoring systems is essential to ensure that efforts effectively reach high-risk groups and help reduce health inequalities across Europe. Moreover, although this review focused exclusively on HES-based measurement approaches, several European countries maintain high-quality national registers with migration-stratified cardiometabolic data. Integrating insights from these register systems with HES findings offers an important avenue for future research and could provide a more comprehensive understanding of cardiometabolic health across diverse populations. Finally, future reviews could extend this work by incorporating behavioral, environmental, socioeconomic, and structural determinants (such as air pollution, traffic noise, neighborhood deprivation, green space availability) alongside physiological risk factors and clinical outcomes to capture the full complexity of cardiometabolic health in migrant-origin groups and ethnic minorities.

Strengths and limitations

A key strength of this study is the use of a transparent and rigorous scoping review methodology, which enabled a systematic and structured overview of HESs targeting persons of migrant origin and ethnic minority groups in Europe. The review process followed established guidelines, ensuring clarity and reproducibility. To maximize comprehensiveness and mitigate publication bias, our search strategy encompassed two major academic databases and an extensive multilingual grey literature search of

national public health websites. Although this grey literature search yielded no eligible reports, it clearly highlighted the scarcity of HES data for migrant origin groups and ethnic minorities, pointing to a substantial gap in publicly accessible health examination information for these populations. It also revealed structural challenges: many documents lacked sufficient methodological detail to verify HES eligibility, did not report usable migrant/ethnicity identifiers, or were non-HES reports (e.g., interview surveys, dashboards) without standardized examination measures. Nonetheless, some limitations should be considered. The retrieval of grey literature remains subject to opaque search engine algorithms. Furthermore, some relevant information and data may exist within national health institutes but remain unpublished or restricted for internal government use.

A further limitation of this review is its exclusive focus on physiological risk factors. Although this approach elucidates underlying biological mechanisms, it potentially narrows the scope of risk assessment by omitting behavioral, environmental, and socioeconomic factors that are critical and modifiable contributors to the development of CVD and T2D. This is particularly relevant for migrant origin populations, who often undergo acculturation-related changes in diet and physical activity that elevate cardiometabolic risk [93]. Environmental and structural factors such as air pollution, traffic-related noise, and neighborhood disadvantage further contribute to this risk and disproportionately affect these groups [94].

Finally, the definition of European countries was based on the EU4Health Programme (2021-2027), which may not fully reflect the diversity of health monitoring practices across the broader European region. Despite these considerations, the review offers a robust and informative overview of current practices and gaps in the monitoring of cardiometabolic risk among migrant origin and ethnic minority groups, and indigenous people.

Conclusion

This scoping review reveals substantial gaps in monitoring CVD and T2D risk factors among migrant origin and ethnic minority groups and indigenous people in EU Member States and associated countries. While several regional HESs exist, the lack of nationally representative data, standardized definitions, and consistent methods hampers reliable cross-country comparisons. The heterogeneity in risk factor assessment and defining target populations complicates data harmonization and limits policy relevance. To support equitable and effective health promotion and disease prevention strategies, improved data availability, methodological alignment, and coordinated European efforts are essential. Developing shared guidelines for the inclusion and assessment of migrant origin and ethnic minority groups and indigenous people in health surveillance systems will be essential to better address the growing burden of CVD and T2D in increasingly diverse European societies.

Declarations

Ethics approval: Not applicable.

Consent to participate: Not applicable.

Consent for publication: Not applicable.

Availability of data and materials: The datasets generated and/or analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request.

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List of abbreviations

BMI: Body mass index

CVD: Cardiovascular disease

DBP: Diastolic blood pressure

EU: European Union

FPG: Fasting plasma glucose

HbA1c: Glycated hemoglobin

HDL: High Density Lipoprotein

HES: health examination survey

IFG: Impaired fasting glucose

IGT: Impaired glucose tolerance

LDL: Low Density Lipoprotein

OGTT: oral glucose tolerance test

SBP: Systolic blood pressure

TC: Total Cholesterol

TG: Triglyceride

T2D: Type two diabetes

WC: waist circumference

Supplementary Information

Additional file 1

Additional file 2

Additional file 3

Additional file 4

Additional file 5

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Table 1: Characteristics of included studies

HES dataset	Author, year	Design	Migrant origin/ ethnic minority group / indigenous people	Migrant status definition	Majority group
Netherlands, Amsterdam (regional)					
The Healthy Life in an Urban Setting (HELIUS)	Agyemang (2015) [26] Bouwmeester (2025) [27] Gazzola (2018) [28] Perini (2018a) [29] Perini (2018b) [30] Perini (2019a) [31] Perini (2019b) [32] Perini (2020a) [33] Perini (2020b) [34] Snijder (2017) [35] Van der Linden (2021) [36] Van Laer (2018) [37] Zethof (2021) [38]	Cohort study	South-Asian Surinamese African Surinamese Ghanaian Turkish Moroccan	First and second- generation migrants Self-identification	Dutch
The Healthy Life in an Urban Setting (HELIUS)	Meeks (2015) [39] Van Apeldoorn (2025) [40]	Cohort study	African Surinamese Ghanaian	First and second- generation migrants	Dutch
The SUNSET study	Admiraal (2014) [41]	Cross- sectional study	South-Asian Surinamese African Surinamese	Self-identification	Dutch
The population- based Amsterdam Health Monitor	Hartog (2018) [42]	Cross- sectional study	Turkish Moroccan	First generation migrants	Dutch
International: Netherlands (Amsterdam), Germany (Berlin), UK (London), Ghana (Ashanti region) (all regional)					
The Research on Obesity and Diabetes among	Agyemang (2016) [43] Beune (2019) [44] Boateng (2017) [45]	Cross- sectional study	Ghanaian in Amsterdam, Berlin, and London	First and second- generation migrants	Ghanaian in rural Ghana

African Migrants (RODAM) study	Chilunga (2019) [46] Commodore-Mensah (2020)[47] Darko (2021) [48] Van der Linden (2019) [49] Van der Linden (2022) [50] Van Nieuwenhuizen (2018) [51]			identified through population registers for migrant groups, foreign embassies and association's lists	and urban Ghana
Sweden, Malmö (regional)					
The population-based MEDIM study	Al-Majdoub (2020) [52] Bennet (2014a) [53] Bennet (2014b) [54] Bennet (2014c) [55] Bennet (2015) [56] Bennet (2016) [57] Dhaher (2023) [58] Pikkemaat (2022) [59]	Cross-sectional study	Iraqi	First generation migrants	Swedish
Sweden, Stockholm (regional)					
Main study population: the cohort of 60-year-old men and women Replication cohort: Malmö Diet and Cancer cohort (MDC)	Carlsson (2014) [60]	Cohort study	Finnish Danish Eastern European South-Western European/ North American Middle Eastern Other	First generation migrants	Swedish
4D Diabetes Project: Screening and Treatment of Prediabetes and Diabetes in Primary Care	Saleh (2024) [61] Taloyan (2023) [62]	Pilot study	Foreign-born: African Middle Eastern Asian	First generation migrant	Swedish

Sweden, Malmö, Skövde, Vara, and Stockholm (regional)					
The MEDIM study The follow-up of the Skaraborg Project The Flemingsberg study The 4-D screening program	Hellgren (2017) [63]	Cross-sectional study	Middle Eastern	First generation migrants	Swedish
Norway, Northern and Middle Norway (regional)					
SAMINOR 1 and/or SAMINOR 2	Naseribafrouei (2018) [64] Naseribafrouei (2019) [65] Siri (2018) [66] Siri (2019) [67]	Cross-sectional study	Sami	Self-identification and speaking a Sami language, or having a parent or grandparent speaking a Sami language	non-Sami
Norway, Oslo (regional)					
The Cohort of Norway (CONOR)	Rabanal (2017) [68]	Cross-sectional study	South Asian	First generation migrant	Norwegian
Finland, Helsinki, Espoo, Vantaa, Turku, Tampere and Vaasa (regional)					
Migrant Health and Wellbeing Study (Maamu)	Etchi (2019) [69] Skogberg (2016) [70] Skogberg (2018) [71] Skogberg (2019) [72]	Cross-sectional study	Russian Somali Kurdish	First generation migrants that resided in Finland for at least 1 year	Finnish
Italy, Veneto region (regional)					
N/A	Dalla Zuanna (2023) [73]	Cohort study	Central-Eastern European Central and Southern American North African Sub-Saharan African Asian	First generation migrants	Italian

Italy, Prato (regional)					
The Chinese in Prato (CHIP)	Modesti (2017a) [74] Modesti (2017b) [75]	Cross-sectional study	Chinese	First generation migrants	Italian
Slovakia, Eastern Slovakia (regional)					
The population-based HepaMeta study	Fedačko (2014a) [76] Fedačko (2014b) [77] Hubková (2014) [78]	Cross-sectional study	Roma	Selected settlements, reached via local Roma community workers	Non-Roma: □ In the vicinity of Roma community □ Outside the area of Roma community
Estonia, Tallin (regional)					
N/A	Kaldmäe (2017) [79]	Cross-sectional study	Russian Other	Unknown	Estonian
Portugal, Universities (regional) Cape Verde, Universities (regional)					
N/A	Tavares (2018) [80]	Cross-sectional study	Cape Verdean	First generation migrants (students) living in Portugal at least 12 months	Cape Verdean in Cape Verde Portuguese Caucasian in Portugal
Bulgaria, South Bulgaria (regional)					
N/A	Delcheva (2023) [81]	Cross-sectional study	Roma	not known	non-Roma
Germany, Würzburg (regional)					
Characteristics and Course of Heart Failure Stages A-B and Determinants of	Morbach (2019) [82]	Cohort Study	EU country Russian Other Did not specify	First generation migrants who speak their native language	German

Progression (STAA B)					
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